We can now solve this system of equations for $\theta_1, \theta_2, \ldots, \theta_p$:

$$\begin{aligned}
\theta_1 &= g_1(X_1, X_2, \ldots, X_n) \\
\theta_2 &= g_2(X_1, X_2, \ldots, X_n) \\
&\vdots \\
\theta_p &= g_p(X_1, X_2, \ldots, X_n)
\end{aligned}$$

These equations are built on the (incorrect) belief that $M_i = E(X^i)$.

In fact, $M_i \approx E(X^i)$, so

$$\theta_i \approx g_i(X_1, X_2, \ldots, X_n)$$

and we construct our estimator as

$$\widehat{\theta}_{i,\text{MOM}} = g_i(X_1, X_2, \ldots, X_n)$$

Exercise: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. from the Exponential(λ) distribution. (Recall that this means that $E(X_i) = 1/\lambda$.) What is the method of moments estimator for λ?

p=1 here, need only one equation

$M_1 = \sum X_i/n = \bar{X}$

$E(X) = 1/\lambda \leftarrow$ first pop. moment

Set equal to each other $M_1 = E(X)$

i.e. $\bar{X} = 1/\lambda$

Solve for parameter: $\lambda = 1/\bar{X}$

So, MOM estimator for λ is $\hat{\lambda}_{mom} = 1/\bar{X}$

Note that $E(\hat{\lambda}_{mom}) \neq 1/E(\bar{X}) = \lambda$, so $\hat{\lambda}_{mom}$ is biased for λ

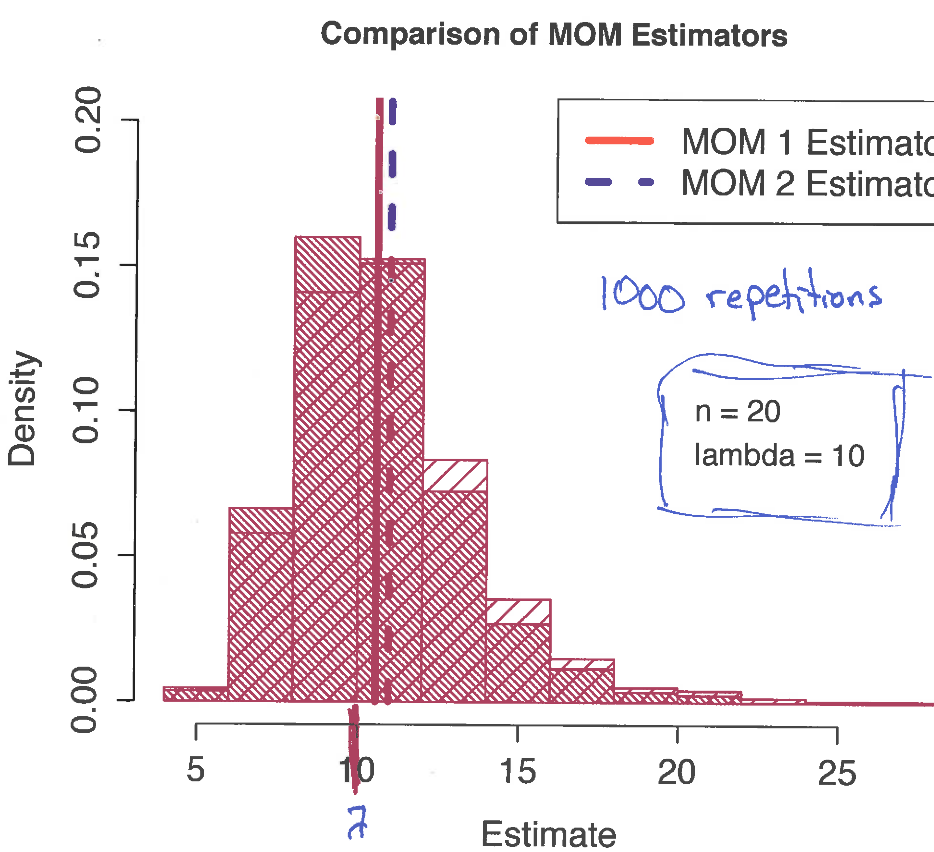
Exercise: What is the result if, instead of the first moment, the second moment is used to construct the estimator for λ in the previous example?

$E(X^2) = V(X) + E(X)^2 = 1/\lambda^2 + 1/\lambda^2$ second pop. moment

$m_2 = \sum X_i^2 / n$ second sample moment

Set equal: $m_2 = E(X^2)$, $\sum X_i^2/n = 2/\lambda^2$

Solve for λ, get $\hat{\lambda}_{mom,2} = \sqrt{\frac{2n}{\sum X_i^2}}$



Exercise: Is the method of moments estimator found above (using the first moment) an unbiased estimator for λ? Can you adjust it to make it an unbiased estimator?

(recall $\sum X_i \sim \text{Gamma}(n, \lambda)$, $\sum X_i / n \sim \text{Gamma}(n, n\lambda)$)

We know: $\bar{X} \sim \text{Gamma}(n, n\lambda)$

So, $1/\bar{x} \sim \text{InvGamma}(n, n\lambda)$

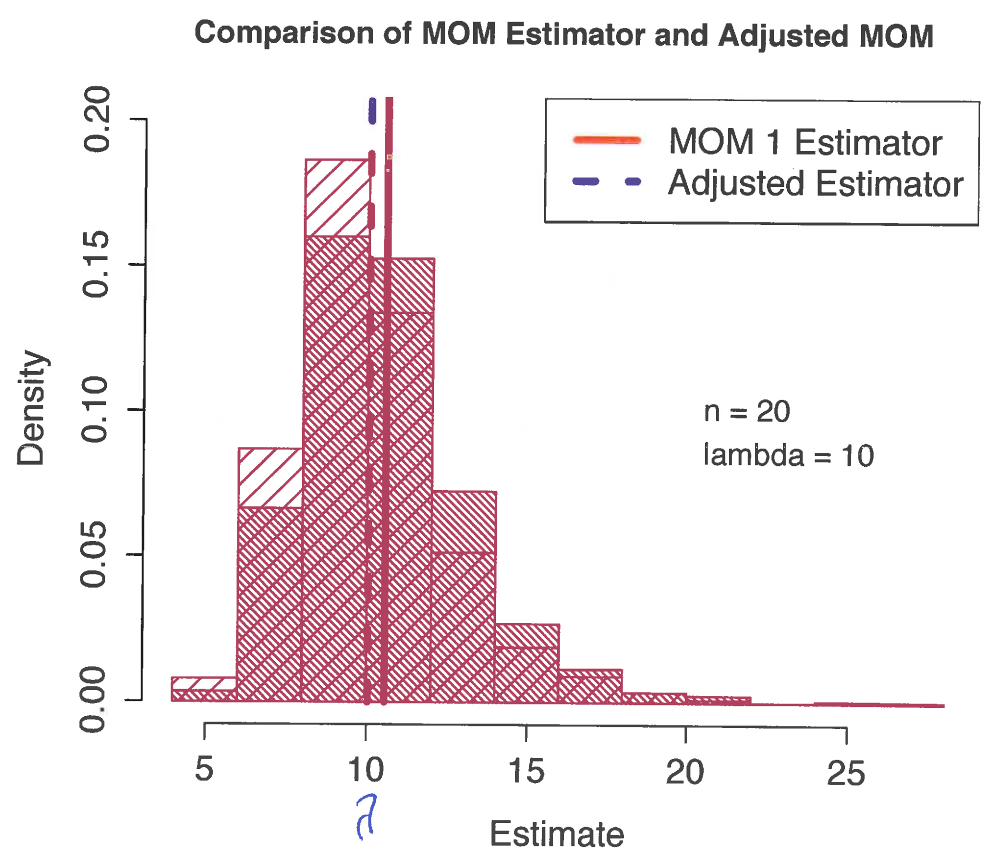
If $Y \sim \text{InvGamma}(\alpha, \beta)$, then $E(Y) = \beta/(\alpha - 1)$, $\alpha > 1$

So, $E(\hat{\lambda}_{mom,1}) = E(1/\bar{x}) = \frac{n\lambda}{n-1} = \left(\frac{n}{n-1}\right)\lambda \neq \lambda$

$\hat{\lambda}_3 = \left(\frac{n-1}{n}\right)(1/\bar{x})$, so $E(\hat{\lambda}_3) = \lambda$

Also, $V(\hat{\lambda}_3) = \left(\frac{n-1}{n}\right)^2 V(1/\bar{x}) = \left(\frac{n-1}{n}\right)^2 V(\hat{\lambda}_{mom,1}) < V(\hat{\lambda}_{mom,1})$

So, $MSE(\hat{\lambda}_3) < MSE(\hat{\lambda}_{mom,1})$



```
# For the comparison of estimators example

# Set the seed for the random number generator, so that the
# results can be replicated

set.seed(0)


n = 20          # The Sample Size
theta = 10      # The true value of theta
reps = 1000     # Number of times to repeat simulation


# Initialize vectors

# Note that there are three estimators here:
# The first MOM estimator, the second MOM estimator,
# and the "adjusted" estimator (denoted theta-hat 3
# in the lecture notes)

mom1 = numeric(reps)
mom2 = numeric(reps)
adju = numeric(reps)

mom1better = 0
adjubetter = 0

for(i in 1:reps)
{

# Simulate the data set

   x = rexp(n, theta)

# Calculate the three estimators

   mom1[i] = 1/mean(x)
   mom2[i] = sqrt(2*n/sum(x^2))
   adju[i] = mom1[i]*(n-1)/n

# Determine if MOM1 is better than MOM2

   if(abs(mom1[i] - theta) < abs(mom2[i] - theta))
   {
      mom1better = mom1better + 1
   }

# Determine if the adjusted estimator is better than MOM1

   if(abs(mom1[i] - theta) > abs(adju[i] - theta))
```

```
    {
      adjubetter = adjubetter + 1
    }
}


# Create the comparison plots

hist(mom1, prob=T, col=2, density=30, angle=-45,
    main="Comparison of MOM Estimators",
    cex.lab=1.3,cex.axis=1.3,xlab="Estimate",ylim=c(0,0.20))
lines(c(mean(mom1),mean(mom1)),c(0,100),col=2,lwd=4)
hist(mom2,prob=T,col=4,density=10,add=T)
lines(c(mean(mom2),mean(mom2)),c(0,100),col=4,lwd=4,lty=2)
legend("topright",col=c(2,4),lty=c(1,2),legend=c("MOM 1
Estimator","MOM 2 Estimator"),cex=1.3,lwd=4)

hist(mom1, prob=T, col=2, density=30, angle=-45,
    main="Comparison of MOM Estimator and Adjusted MOM",
    cex.lab=1.3,cex.axis=1.3,xlab="Estimate",ylim=c(0,0.20))
lines(c(mean(mom1),mean(mom1)),c(0,100),col=2,lwd=4)
hist(adju,prob=T,col=4,density=10,add=T)
lines(c(mean(adju),mean(adju)),c(0,100),col=4,lwd=4,lty=2)
legend("topright",col=c(2,4),lty=c(1,2),legend=c("MOM 1
Estimator","Adjusted Estimator"),cex=1.3,lwd=4)


# Print the means

print(c(mean(mom1), mean(mom2), mean(adju)))


# Print the variances

print(c(var(mom1), var(mom2), var(adju)))
```

The comparisons made above between the three estimators are based **entirely on simulations**. There is **no** observed data involved at all. This is the frequentist approach to assessing estimators: **how does the estimator perform in repeated use?**.

As seen above, simulations are a useful tool for cases where the mean and variance derivations are not available.

The full sequence of R commands to perform the simulations can be found on Canvas. For each estimator, 1000 samples of size $n = 20$ are generated under the assumption that $\lambda = 10$.

The simulations provide us 1000 estimates using each estimator. These are stored in the objects `mom1`, `mom2`, and `adju`.

Exercise: Interpret the output below. What is the approximate MSE of each of the three estimators?

```
> print(c(mean(mom1), mean(mom2), mean(adju)))
[1] 10.56553 10.96759 10.03725

> print(c(var(mom1), var(mom2), var(adju)))
[1] 6.574420 8.224773 5.933414
```

MSE = $\text{bias}^2 + \text{Var}$

MSE of $\hat{\lambda}_{mom,1} = (10.57-10)^2 + 6.57 = 6.89$

$\hat{\lambda}_{mom,2} = (10.97-10)^2 + 8.22$

$\hat{\lambda}_{adju} = 0^2 + 5.93 = 5.93$

Important Cases

There are a few estimation situations that worth special attention.

Case 1: The first we have already explored, that of estimating a population proportion. We found that the sample proportion is unbiased and has a standard error of

$$\sqrt{\frac{p(1-p)}{n}}$$

Of course, we do not know the value of p, so typically it is replaced by its best estimate and the approximate SE is reported as

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Exercise: We have derived the mean and variance of the sample proportion, but what is its distribution?

$\hat{p}$ is approx. normal by CLT

$\hat{p} = \sum X_i / n$ where $X_i = \begin{cases} 1, & \text{when } i^{th} \text{ trial is "success"} \\ 0, & \text{" " " " "failure"} \end{cases}$

$X_1, X_2, .., X_n$ are iid binomial$(1, p)$

The approx. improves as n grows.

Rule of thumb: Approx. is fine if $np \geq 5$ and $n(1-p) \geq 5$

Case 2: Suppose that $X_1, X_2, \ldots, X_n$ are random variables and $E(X_i) = \mu$ and $V(X_i) = \sigma^2$ $<\infty$ for all i. Further assume that these random variables are uncorrelated (meaning that every pair of random variables is uncorrelated).

Then, the standard estimator for μ is the sample mean, $\overline{X}$, and the standard estimator for σ^2 is the sample variance,

$$s^2 = \sum_{i=1}^{n} \left(X_i - \overline{X}\right)^2 \Big/ (n-1)$$

Note that the case where $X_1, X_2, \ldots, X_n$ are i.i.d. is a special instance of this situation.

Exercise: Derive the mean, variance, and standard error of the sample mean. What is its MSE as an estimator of μ?

$E(\bar{X}) = E(\frac{1}{n}\sum X_i) = \frac{1}{n}\sum E(X_i) = \mu$

$\bar{X}$ is an unbiased estimator of μ

$V(\bar{X}) = V(\frac{1}{n}\sum X_i) = \sigma^2/n$

MSE of $\bar{X}$ (as an estimator μ) $= \sigma^2/n$

SE of $\bar{X}$ is $\sigma/\sqrt{n}$

Often reported as $s/\sqrt{n}$

Exercise: If we did assume that $X_1, X_2, \ldots, X_n$ are i.i.d., what could we say about the distribution of the sample mean?

As long as n is "sufficiently" large, then CLT says $\bar{X}$ is approx. normal

Specifically, $\bar{X} \approx N(\mu, \sigma^2/n)$

Exercise: As defined, the sample variance is an unbiased estimator of σ^2. (The proof is not difficult, but a little tedious.)

Consider the following question: What choice of a minimizes the quantity

$$\sum_{i=1}^{n} (X_i - a)^2 \ ?$$

How does this result suggest it would be a good idea to estimate σ^2 by dividing the sum of squares by a constant less than n?

$\Sigma(X_i - a)^2$ is minimized when $a = \bar{X}$

$\frac{1}{n}\Sigma(X_i - \mu)^2$ is an unbiased estimator of σ^2

since $E((X_i - \mu)^2) = \sigma^2$

$$\frac{1}{n}\sum(X_i-\bar{X})^2 \le \frac{1}{n}\sum(X_i-\mu)^2$$

If I use $\frac{1}{n}\sum(X_i-\bar{X})^2$ as my estimator, I will (on average) underestimate σ^2

Would be better to use estimator of the form $\frac{1}{k}\sum(X_i-\bar{X})^2$ where $k<n$

Turns out $k=n-1$ is unbiased for σ^2

Case 3: Suppose that $X_1, X_2, \ldots, X_n$ are i.i.d. with the normal distribution with mean μ and variance σ^2.

Of course, this is a special instance of "Case 2" presented above, and the sample mean and sample variance remain unbiased estimators of μ and σ^2, respectively.

We can make some stronger statements, as well.

First, we know that $\overline{X}$ has **exactly** the normal distribution with mean μ and variance σ^2/n for **any** sample size.

Further results depend on properties of probability distributions that we will present now.

The Chi-squared distribution with k degrees of freedom is a special case of the Gamma distribution. In particular, if X has the Gamma$(k/2, 1/2)$ distribution, then X has the Chi-square distribution with k degrees of freedom.

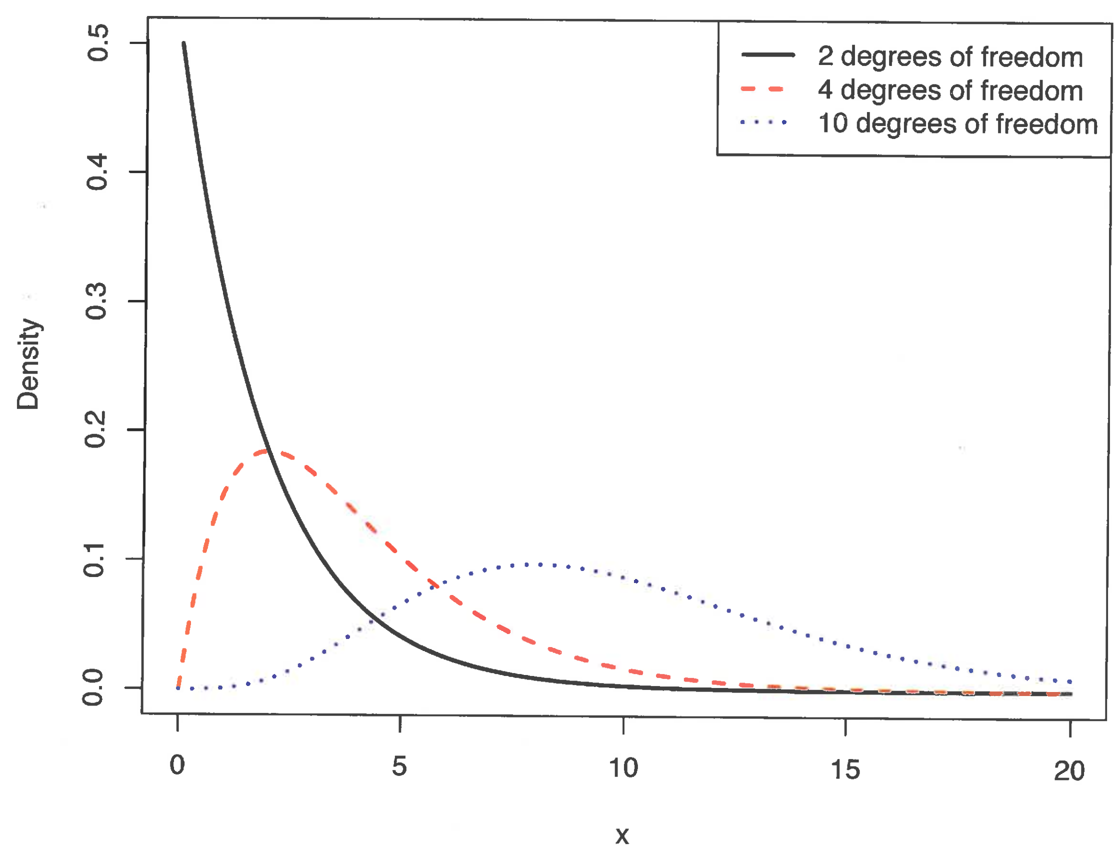
Also, if $Z_1, Z_2, \ldots, Z_k$ are i.i.d. standard normal random variables, then

$$X = \sum_{i=1}^{k} Z_i^2$$

has the Chi-squared distribution with k degrees of freedom.

If X has this distribution, then $E(X) = k$ and $V(X) = 2k$.

Plots of pdfs from the Chi-squared distribution are shown on the following slide.



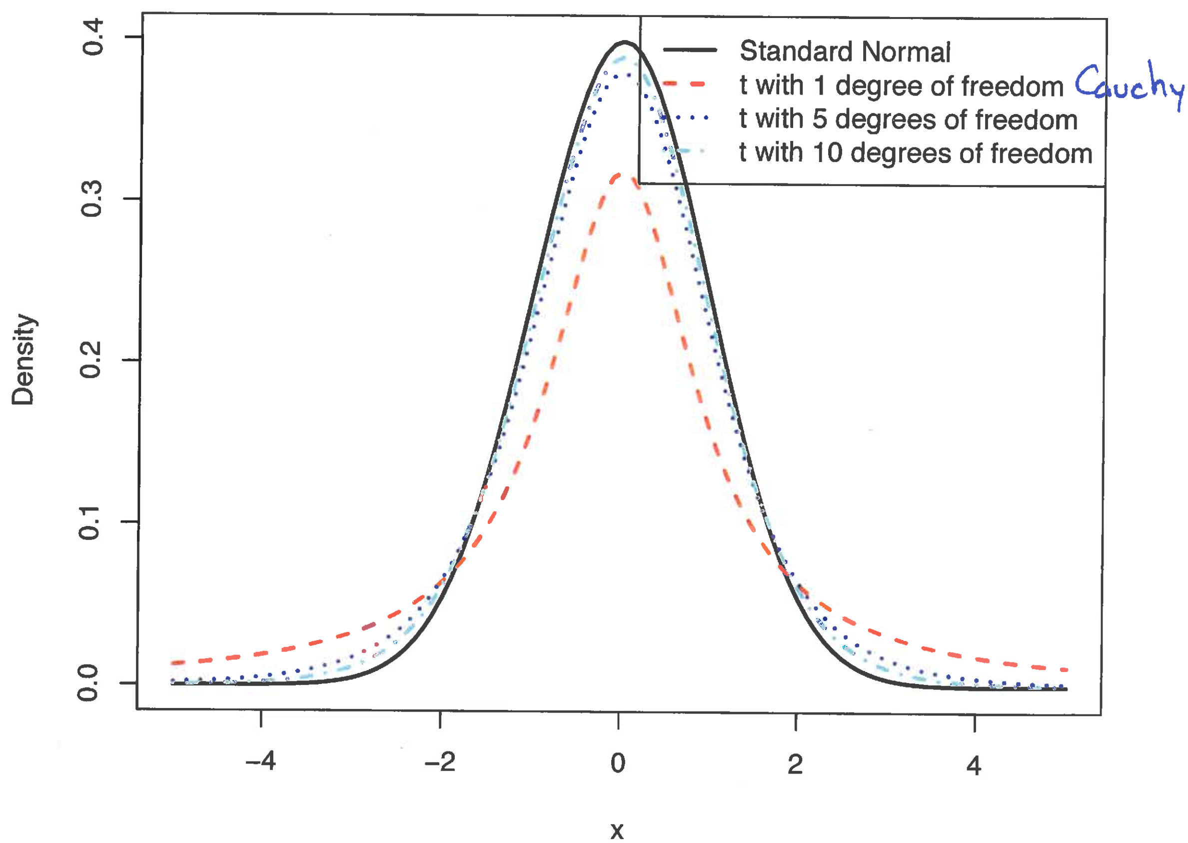
The *t-distribution with ν degrees of freedom* is defined to be a random variable T that can be constructed as

$$T = \frac{Z}{\sqrt{X/\nu}}$$

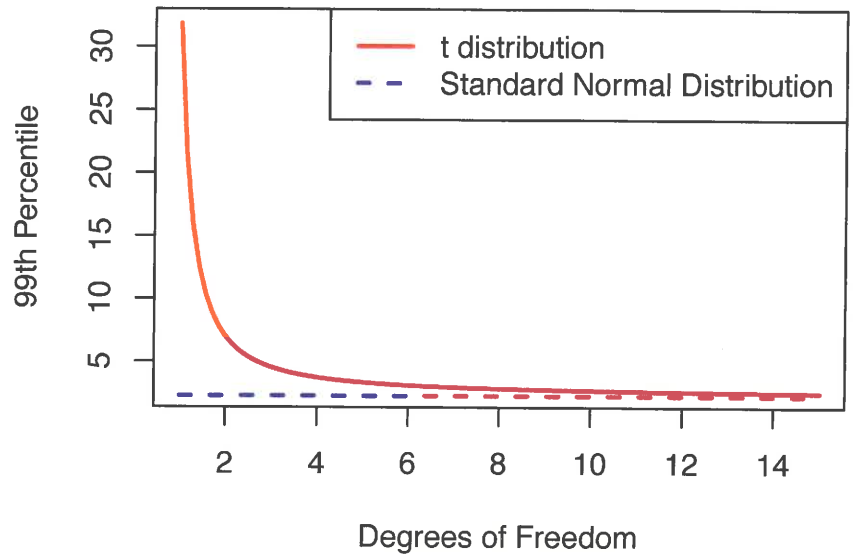
where Z is a random variable with the standard normal distribution, and X is a random variable with the Chi-squared distribution with ν degrees of freedom. Further, Z and X are assumed independent.

This distribution has a "normal-shaped" pdf, but has heavier tails than the normal distribution. As ν increases, the distribution converges to the standard normal distribution.

It holds that $E(T) = 0$ when $\nu \geq 2$ and that $V(T) = \nu/(\nu - 2)$ when $\nu \geq 3$.



The figure below makes clearer how much additional probability is in the tails of the t-distribution compared to the standard normal. For $\nu = 1$, 1% of the observations will be larger than 30. The 99th percentile of the standard normal distribution is 2.33.



Now we can state the four **classic results** that hold under our "Case 3:"

Assuming that $X_1, X_2, \ldots, X_n$ are i.i.d. Normal(μ, σ^2), then

1. $\overline{X}$ and s^2 are independent.
2. The quantity
$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$
has the standard normal distribution.
3. The quantity
$$\frac{(n-1)s^2}{\sigma^2}$$
has the Chi-squared distribution with $n-1$ degrees of freedom.
4. The quantity
$$\frac{\overline{X} - \mu}{s/\sqrt{n}}$$
has the t-distribution with $n-1$ degrees of freedom.

46-923: Supplement: Proofs

These proofs will not be covered on the homeworks or the exam, but it is important to know the claims.

Throughout assume that $X_1, X_2, \ldots, X_n$ are i.i.d. from the normal distribution with mean μ and variance σ^2.

Claim 1: $\overline{X}$ and S^2 are independent.

Proof: First, note that since $X_1, X_2, \ldots, X_n$ are independent and normal, then the random vector $\mathbf{X}$ consisting of $X_1, X_2, \ldots, X_n$ is multivariate normal with covariance matrix $\mathbf{\Sigma} = \sigma^2\mathbf{I}$. (To see this, derive the joint pdf of $\mathbf{X}$ by taking the product of the pdfs of X_1, X_2, ..., X_n and note that its form matches the multivariate normal with covariance matrix $\mathbf{\Sigma}$.)

Fix any $i \in \{1, 2, \ldots, n\}$. We see that we can write

$$\begin{bmatrix} \overline{X} \\ (X_i - \overline{X}) \end{bmatrix} = \mathbf{A}_i\mathbf{X}$$

with

$$\mathbf{A}_i = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \frac{1}{n} & \cdots & \frac{1}{n} \\ -\frac{1}{n} & \cdots & -\frac{1}{n} & 1 - \frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} \end{bmatrix}$$

where the $1 - 1/n$ entry in the second row of $\mathbf{A}_i$ is in the i^{th} column. This implies that

$$\begin{bmatrix} \overline{X} \\ (X_i - \overline{X}) \end{bmatrix}$$

is bivariate normal with covariance matrix $\mathbf{A}_i\mathbf{\Sigma}\mathbf{A}_i^T = \sigma^2\mathbf{A}_i\mathbf{A}_i^T$. Finally, the off-diagonal entry of $\sigma^2\mathbf{A}_i\mathbf{A}_i^T$ is

$$\sigma^2\left[\left(\frac{1}{n}\right)\left(1 - \frac{1}{n}\right) + \sum_{j \neq i}\left(\frac{1}{n}\right)\left(-\frac{1}{n}\right)\right] = 0.$$

Of course, this off-diagonal element is the covariance between $\overline{X}$ and $(X_i - \overline{X})$. The conclusion is that $\overline{X}$ and $(X_i - \overline{X})$ are uncorrelated and hence independent.

Next, note that the sample variance S^2 can be written as a function of $(X_i - \overline{X})$ for $i = 1, 2, \ldots, n$:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \overline{X}\right)^2.$$

Hence, S^2 is a function of random variables, each of which is independent of $\overline{X}$. This implies that $\overline{X}$ is independent of S^2. □

Claim 2: The quantity

$$\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$$

has the standard normal distribution.

Proof: We showed this in probability. □

Claim 3: The quantity

$$\frac{(n-1)\,S^2}{\sigma^2}$$

has the chi-squared distribution with $n-1$ degrees of freedom.

Proof: To start, note that

$$\begin{aligned}
\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2 &= \sum_{i=1}^{n}\left(\frac{X_i-\overline{X}+\overline{X}-\mu}{\sigma}\right)^2 \\
&= \frac{1}{\sigma^2}\sum_{i=1}^{n}\left[\left(X_i-\overline{X}\right)^2+2\left(X_i-\overline{X}\right)\left(\overline{X}-\mu\right)+\left(\overline{X}-\mu\right)^2\right] \\
&= \frac{1}{\sigma^2}\left[\sum_{i=1}^{n}\left[\left(X_i-\overline{X}\right)^2\right]+n\left(\overline{X}-\mu\right)^2\right] \quad \text{since } \textstyle\sum_i (X_i-\overline{X})=0 \\
&= \sum_{i=1}^{n}\left(\frac{X_i-\overline{X}}{\sigma}\right)^2+\left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\right)^2 \\
&= W_2+W_3
\end{aligned}$$

where

$$W_2=\sum_{i=1}^{n}\left(\frac{X_i-\overline{X}}{\sigma}\right)^2 \quad \text{and} \quad W_3=\left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\right)^2.$$

Also define

$$W_1=\sum_{i=1}^{n}\left(\frac{X_i-\mu}{\sigma}\right)^2$$

so that we have shown $W_1=W_2+W_3$.

We know that W_1 has the chi-squared distribution with n degrees of freedom and that W_3 has the chi-squared distribution with a single degree of freedom. Note that the only random variable W_2 depends on is S^2 and the only random variable W_3 depends on is $\overline{X}$. Since S^2 and $\overline{X}$ are independent, it follows that W_2 and W_3 are independent. Hence,

$$M_{W_1}(t) = M_{W_2}(t)M_{W_3}(t)$$

and

$$\begin{aligned} M_{W_2}(t) &= M_{W_1}(t)\Big/M_{W_2}(t) \\ &= \left(\frac{1}{1-2t}\right)^{n/2}\Big/\left(\frac{1}{1-2t}\right)^{1/2} \\ &= \left(\frac{1}{1-2t}\right)^{\left(\frac{n-1}{2}\right)}. \end{aligned}$$

So, by the uniqueness of the MGF, W_2 has the chi-squared distribution with $n-1$ degrees of freedom. Finally,

$$W_2 = \sum_{i=1}^{n}\left(\frac{X_i - \overline{X}}{\sigma}\right)^2 = \frac{(n-1)\,S^2}{\sigma^2}. \quad \square$$

Claim 4: The quantity

$$\frac{\overline{X}-\mu}{s/\sqrt{n}}$$

has the t-distribution with $n-1$ degrees of freedom.

Proof: Start by defining random variables

$$Z = \left(\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}\right) \quad \text{and} \quad V = \frac{(n-1)\,S^2}{\sigma^2}.$$

Based on the above results, we know that Z has the standard normal distribution and V has the chi-squared distribution with $n-1$ degrees of freedom. We also know that Z and V are independent.

It is not difficult to see that

$$\frac{\overline{X}-\mu}{s/\sqrt{n}} = \frac{Z}{\sqrt{V\Big/(n-1)}}$$

and thus by the definition of the t-distribution, the claim follows. $\square$

Exercise: Under "Case 3" it is easier to show that s^2 is an unbiased estimator of σ^2, and its also easy to derive the variance of this estimator. Do these now.

$$E(s^2) = E\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)s^2\right) = \frac{\sigma^2}{n-1} E\left(\frac{n-1}{\sigma^2} s^2\right) = \left(\frac{\sigma^2}{n-1}\right)(n-1) = \sigma^2$$

$$V(s^2) = V\left(\left(\frac{\sigma^2}{n-1}\right)\left(\frac{n-1}{\sigma^2}\right)s^2\right) = \frac{\sigma^4}{(n-1)^2} V\left(\frac{n-1}{\sigma^2} s^2\right) = 2\sigma^4/(n-1)$$

So, SE of s^2 is $\sqrt{\dfrac{2\sigma^4}{(n-1)}}$

Confidence Intervals

The already-mentioned standard error (SE) is a useful tool for quantifying error in an estimate, but a common alternative is the use of confidence intervals for unknown parameters.

Formally, we call (L, U) a $100(1-\alpha)\%$ confidence interval for θ if

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Hence, if want a "95% confidence interval," set $\alpha = 0.05$. Other common choices are 90% and 99% confidence intervals ($\alpha = 0.10$ and $\alpha = 0.01$, respectively).